

IIITB: The flag bearer

IIITB has contributed significantly to and benefited from the open source ecosystem. Here are its key contributions, as recounted by Prof. Sadagopan.

Open source education

S. Nagarajan, a distinguished engineer from HP, teaches as an adjunct faculty at IITB. His advanced operating system course encourages students to delve into various aspects of open source, including developing device drivers. Some of the student projects under his guidance have significantly impacted the field of open source. With his efforts, a Debian event was hosted in India and a Debian expert group has been formed.

Alongside contributing to open source projects, IIITB has made its courses open, and is also involved in the OpenCourseWare movement.

Modular Open Source Identity Program (MOSIP)

On December 21, 2017, representatives from the Indian Software Products Industry Round Table (iSPIRT) came to IIITB with a request from several countries to create an ID system similar to Aadhaar. They came to us seeking connections with other institutes like IITs, but the conversation led to the realisation that IIITB could contribute to this project. My experience with the Unicode forum and connections in Berkeley and MIT could be leveraged. So, with backing from the Tata Trust and the Bill & Melinda Gates Foundation, and in conformance with the FCRA (Foreign Contribution Regulation Act), we started the project in Bengaluru. This was the first time an academic institute in India had embarked on a project of this scale and nature. The project was named Modular Open Source Identity Program (MOSIP), which is now an international initiative. Along with software and documentation, it also focuses on ecosystem development, involving experts like Krishnan Rajivopalan in standards development. Among the nine countries that have signed up, at least 3-4 have already completed their implementations. Over a hundred billion IDs have been generated.

eGov Foundation

Funded by Nandan Nilekani, operated by Shrikanth Nathamuni, and advocated by me, it has been active for 20 years. The foundation is based in Bengaluru and all the software solutions we develop are open source. Seventeen states in India are using its solutions to deliver various municipal services.

Digit for tax collection

Paying BBMP (Bruhat Bengaluru Mahanagara Palike) tax now takes just about three-and-a-half minutes. Using Digit, BBMP has been able to collect revenues in billions of rupees, with 20 other towns adopting this system.

DIVOC

DIVOC is a software tool that facilitated the entire planning and execution of the Covid vaccination program, including managing and issuing certificates. Thanks to open source, its first version was developed in just a few days, a feat that would typically take weeks, months, or even years.

Miles to go...

India is gradually making significant strides in various sectors, including education and healthcare, by leveraging open source solutions, but there are miles to go.

A critical area being explored is the open sourcing of academic assessment processes. In a country with 1.4 billion people, any large-scale assessment like the JEE exams presents logistical challenges. The aim is to streamline processes such as exam scheduling, registration, distribution of hall tickets, and more, using open source solutions. This approach could revolutionise how large-scale academic assessments are conducted.

In healthcare, following the success in vaccine distribution, we are looking at extending open source methodologies to manage health records and other tests. The Indian government's initiatives in health management are also embracing these approaches.

In the area of land and property management, land records across cities can be standardised by integrating them into systems like DigiLocker, ensuring easy and secure access to property documents.

Despite these advancements, we've only begun to scratch the surface of what open source can achieve in India. We've seen only 5 to 10 per

cent of its true potential. The journey ahead, covering the remaining 90 to 95 per cent, is still long and filled with opportunities.

The time is ripe for action, so every developer should pledge to contribute to open source through coding, documentation, community support, or any other value addition to the open source ecosystem. This is not just a call to engage with technology, it's an invitation to be part of a transformative movement where India becomes an 'uploading nation'. **END** 🐧

This story has been written by Shivangi Kharoo, technical journalist at EFY, based on Prof. S Sadagopan's speech at Open Source India 2023.